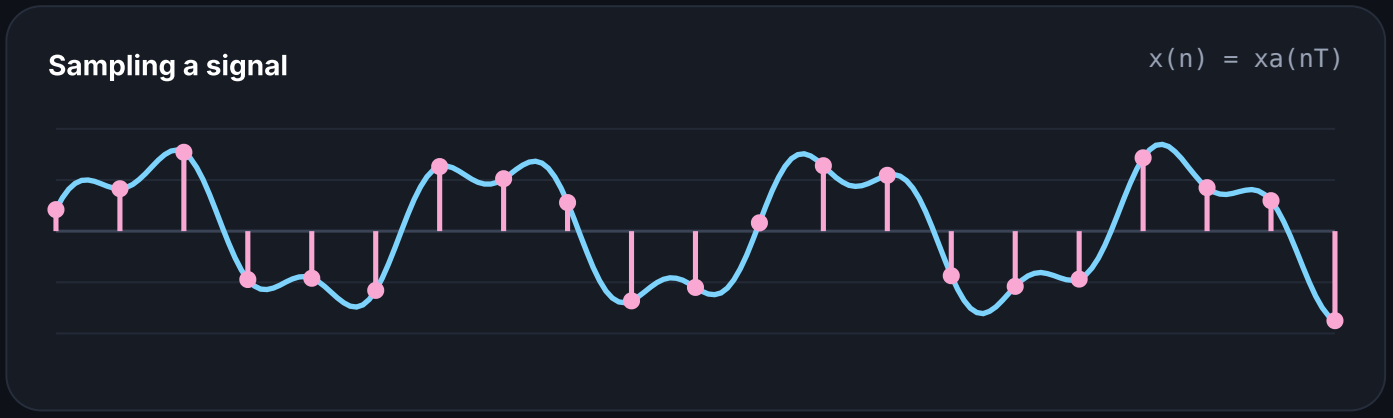
المسائل و Sheet 2
Chapter 2

Digital Signal Processing · Problems, Sheet 2 & Assignment · 15 questions



DFT · IDFT · Parseval · FFT · Butterfly · IFFT

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Lecturer: Dr. Samir M. Kamal

Modern Academy for Engineering & Technology

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المحتوى

Problems, Sheet 2 & Assignment · 15 questions

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إزاي تذاكر من الملف

المسائل خطوة بخطوة بنفس طريقة الدكتور: أمثلة المحاضرات، الامتحانات القديمة، والشيت والأسايمنت. تحت كل مسألة مربع النتيجة، وفي آخر كل جزء صفحة بكل النتايج.

SECTION 01

مسائل المحاضرات

Examples 2.1 لحد 2.4

الأسئلة

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p. 7	Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j}	P2 ☐
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P1

Example 2.1 — DFT of {1, 0, 0, 1}

The DFT of the sequence {1, 0, 0, 1} (sampled at 8 kHz) will be evaluated. Find X(k), plot |X(k)| and Φ(k), and find the Fourier transform components.

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 0 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(3\pi/2) - j\, \sin(3\pi/2)] \\ &= 1 + j \\ &= \mathbf{1 + j} \end{aligned}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi} \\ &= 1 + [\cos(3\pi) - j\, \sin(3\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$\begin{aligned} X(3) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4} \\ &= 1 + 0 + 0 + e^{-j9\pi/2} \end{aligned}$$

$$= 1 + [\cos(9\pi/2) - j\ \sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

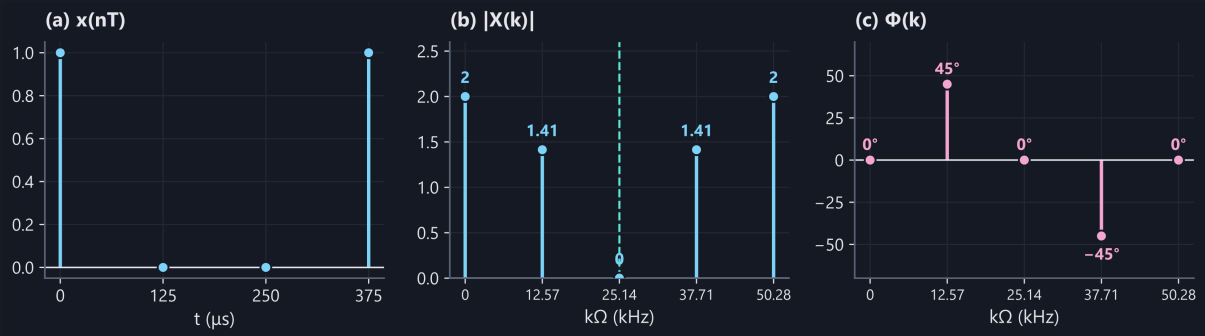
Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
|X(k)|	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

$T = 1/(8\times10^{3}) = 125\ \mu s \quad \Omega = 2\pi/NT = 2\pi/(4 \times 125\times10^{-6}) = 12.57\ k \rightarrow 2\Omega = 25.14\ k,\ 3\Omega = 37.71\ k$

The Fourier transform components are obtained by multiplying the DFT components by T = 125 μs (FT = DFT × T):

$F(0) = 250\ \mu V\ Hz^{-1}, \quad F(\Omega) = (125 + j125)\ \mu V\ Hz^{-1}, \quad F(2\Omega) = 0, \quad F(3\Omega) = (125 - j125)\ \mu V\ Hz^{-1}$



النتيجة $X(k) = \{2,\ 1 + j,\ 0,\ 1 - j\} \cdot |X(k)| = \{2,\ 1.414,\ 0,\ 1.414\} \cdot \Phi(k) = \{0°,\ 45°,\ 0°,\ -45°\}$

المصدر — من الـ PDFs بتاعتك

مسائل المحاضرات

LEC 5 · Example 2.1 · p. 11 — المحاضرة

1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-5-Ch-2.pdf

P2

Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j}

Derive the time series {1, 0, 0, 1} from its DFT components {2, 1 + j, 0, 1 − j}.

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi/2} + 0 + (1 - j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi} + 0 + (1 - j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

مسائل المحاضرات

النتيجة

```
x(n) = {1, 0, 0, 1} (the original sequence)
```

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 5 · Example 2.2 · p. 23

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P3

Example 2.3 — FFT of {1, 0, 0, 1}

Obtain the DFT of {1, 0, 0, 1} by means of the decimation-in-time FFT algorithm.

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

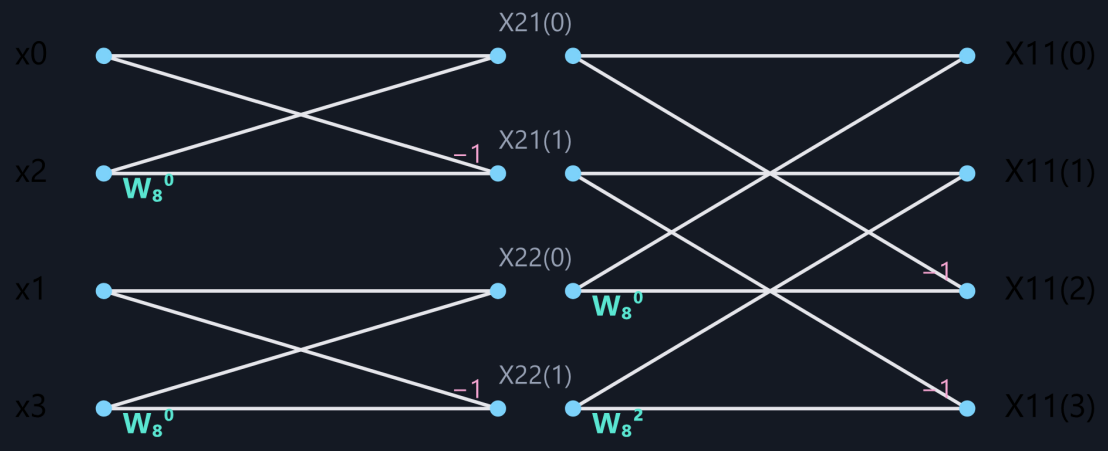
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained in Section 2.1, but were more readily obtained using the FFT algorithm.

مسائل المحاضرات

النتيجة

`X(k) = {2, 1 + j, 0, 1 − j} — same as the DFT of Example 2.1`

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 6 · Example 2.3 · p. 16

`1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-6-Ch-2.pdf`

P4

Example 2.4 — FFT of {0.5, 1, 1, 0.5}, sampled at 125 Hz

The first four sampled voltage values of a 10 Hz bandwidth signal sampled at 125 Hz were (0.5, 1, 1, 0.5). Obtain the DFT using the FFT, and hence the Fourier transform.

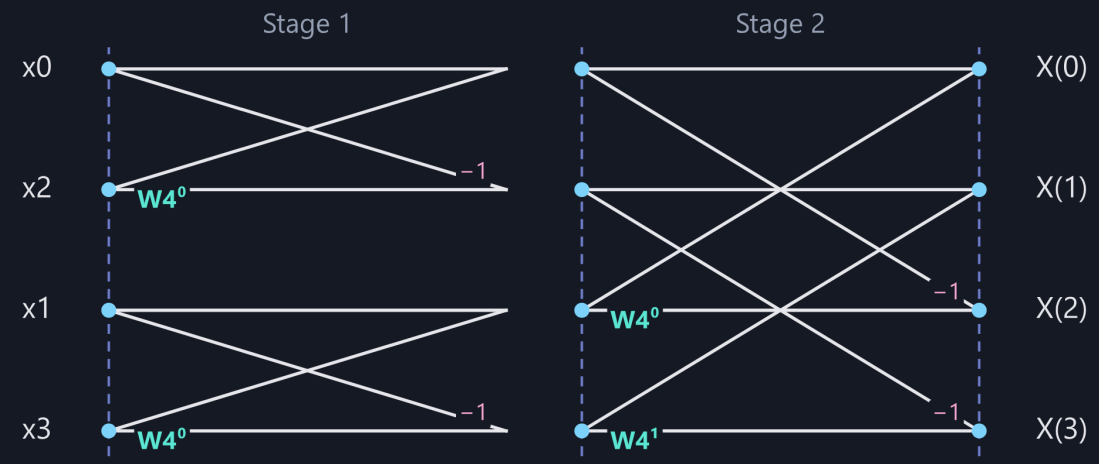
Re-ordered sequence: x(0), x(2), x(1), x(3). $W_4^0 = 1$, $W_4^1 = e^{-j2\pi/4} = -j$

$$G(0) = x(0) + x(2) = 1.5 \qquad G(1) = x(0) - x(2) = -0.5$$
$$H(0) = x(1) + x(3) = 1.5 \qquad H(1) = x(1) - x(3) = 0.5$$

$$X(0) = G(0) + W_4^0 H(0) = 1.5 + 1.5 = \mathbf{3}$$
$$X(1) = G(1) + W_4^1 H(1) = -0.5 - j(0.5) = \mathbf{-0.5 - j0.5}$$
$$X(2) = G(0) - W_4^0 H(0) = 1.5 - 1.5 = \mathbf{0}$$
$$X(3) = G(1) - W_4^1 H(1) = -0.5 + j(0.5) = \mathbf{-0.5 + j0.5}$$

∴ X(k) = **{3, −0.5 − j0.5, 0, −0.5 + j0.5}** (same as the DFT).

4-point DIT FFT flowgraph (Fig. 2.4)



$T = 1/125 = 0.008$ s; the signal period is $1/10 = 0.1$ s → $T/\text{period} = 0.08 \ll 1$

→ FT = T × DFT

FT = {0.024, −0.004(1 + j), 0, 0.004(−1 + j)}

مسائل المحاضرات

النتيجة

$X(k) = \{3, -0.5 - j0.5, 0, -0.5 + j0.5\}$ · $FT = \{0.024, -0.004(1 + j), 0, 0.004(-1 + j)\}$

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 6 · Example 2.4 · p. 22

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★ Cheat sheet · مسائل المحاضرات

النتايج النهائية

$X(k) = \{2, 1 + j, 0, 1 - j\}$ · $\lvert X(k)\rvert = \{2, 1.414, 0, 1.414\}$ · $\Phi(k) = \{0°, 45°, 0°, -45°\}$	P1
$x(n) = \{1, 0, 0, 1\}$ (the original sequence)	P2
$X(k) = \{2, 1 + j, 0, 1 - j\}$ — same as the DFT of Example 2.1	P3
$X(k) = \{3, -0.5 - j0.5, 0, -0.5 + j0.5\}$ · $FT = \{0.024, -0.004(1 + j), 0, 0.004(-1 + j)\}$	P4

SECTION 02

مسائل الامتحانات القديمة

الأسئلة	
p. 14 x(n) = {1, 0, 1, 1}	P9 ☐
p. 19 x(n) = {1, 0, 1, 0}	P10 ☐
p. 24 x(n) = {1, 1, 0, 1}	P11 ☐
p. 29 x(n) = {1, 0, 0, 1}	P12 ☐

مسائل الامتحانات القديمة

P9 متكرر ●10×

x(n) = {1, 0, 1, 1}

For the sequence x(n) = {1, 0, 1, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 1 \\ &= \mathbf{3} \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + e^{-j\pi} + e^{-j3\pi/2} \\ &= 1 + [\cos(\pi) - j\,\sin(\pi)] + [\cos(3\pi/2) - j\,\sin(3\pi/2)] \\ &= 1 + (-1) + j \\ &= \mathbf{j} \end{aligned}$$

Thus X(1) = j, of magnitude 1 and phase angle Φ(1) = 90°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + e^{-j2\pi} + e^{-j3\pi} \\ &= 1 + [\cos(2\pi) - j\,\sin(2\pi)] + [\cos(3\pi) - j\,\sin(3\pi)] \\ &= 1 + 1 + (-1) \\ &= \mathbf{1} \end{aligned}$$

Thus X(2) = 1, of magnitude 1 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)] + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + (-1) + (-j)$$
$$= \mathbf{-j}$$

Thus X(3) = –j, of magnitude 1 and phase angle Φ(3) = −90°.

So, the time series x(n) = {1, 0, 1, 1} has the DFT given by **X(k) = {3, j, 1, –j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 1, and

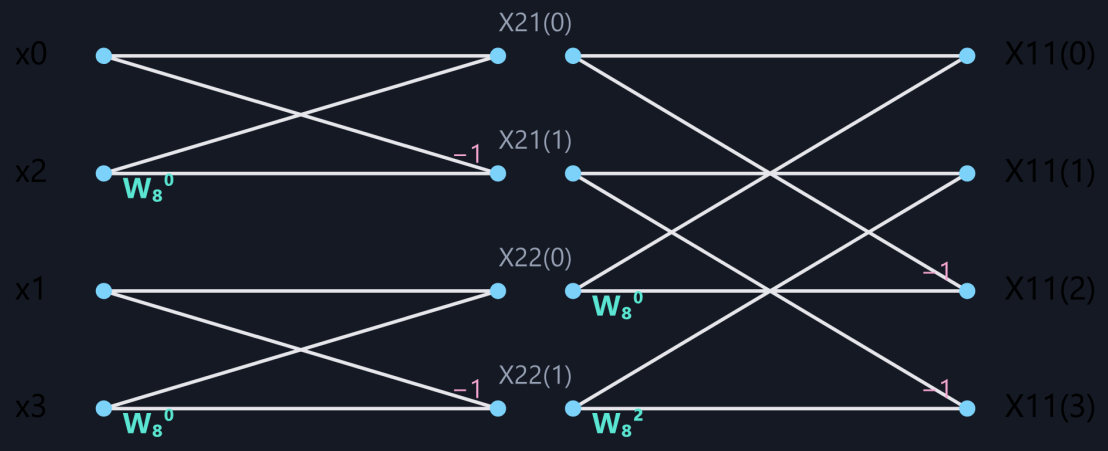
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 1 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(-1) = 0 - j(-1) = \mathbf{j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 1 = \mathbf{1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(-1) = 0 + j(-1) = \mathbf{-j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 1, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi/2} + 1\ e^{j\pi} + (-j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(j) + 1(-1) + (-j)(-j)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi} + 1\ e^{j2\pi} + (-j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + j(-1) + 1(1) + (-j)(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j3\pi/2} + 1\ e^{j3\pi} + (-j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(-j) + 1(-1) + (-j)(j)]$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)|$ = {3, 1, 1, 1}

$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 1^2 = \mathbf{3}$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	j	1	−j
$\|X(k)\|$	3	1	1	1
Φ(k)	0°	90°	0°	−90°

Spectra of {1, 0, 1, 1}



مسائل الامتحانات القديمة

(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

النتيجة `X(k) = {3, j, 1, −j} · IDFT → {1, 0, 1, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.`

Key words (iii) Parseval's theorem

المصدر — من الـ PDFs بتاعتك

المحاضرة

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المحاضرة

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المحاضرة

LEC 5 · Sec. 2.3 · p. 27
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المحاضرة

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جه في

Final 20-1-2025 · Q4-c | Final 23-1-2023 · Q3-b | Final 17-8-2026 · Q2-b | Final 18-1-2024 · Q2-c | Final 12-1-2019 · Q2-c | Final 2-1-2017 · Q2-c | Midterm 31-7-2023 · Q2-b | Midterm 8-12-2020 · Q2-c | Midterm 2 · Q2-c | Book Assignment A2.2

5-Exams/Final/ · 5-Exams/Midterm/ · 7-Books (References)/Extract Question From Book/

P10 متكرر ●4×

x(n) = {1, 0, 1, 0}

For the sequence x(n) = {1, 0, 1, 0}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 0 \\ &= 2 \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + e^{-j\pi} + 0 \\ &= 1 + [\cos(\pi) - j\, \sin(\pi)] \\ &= 1 + (-1) \\ &= 0 \end{aligned}$$

Thus X(1) = 0, of magnitude 0 and phase angle Φ(1) = 0°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + e^{-j2\pi} + 0 \\ &= 1 + [\cos(2\pi) - j\, \sin(2\pi)] \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

Thus X(2) = 2, of magnitude 2 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + 0$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + (-1)$$
$$= 0$$

Thus X(3) = 0, of magnitude 0 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 0, 1, 0} has the DFT given by **X(k) = {2, 0, 2, 0}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 0, and

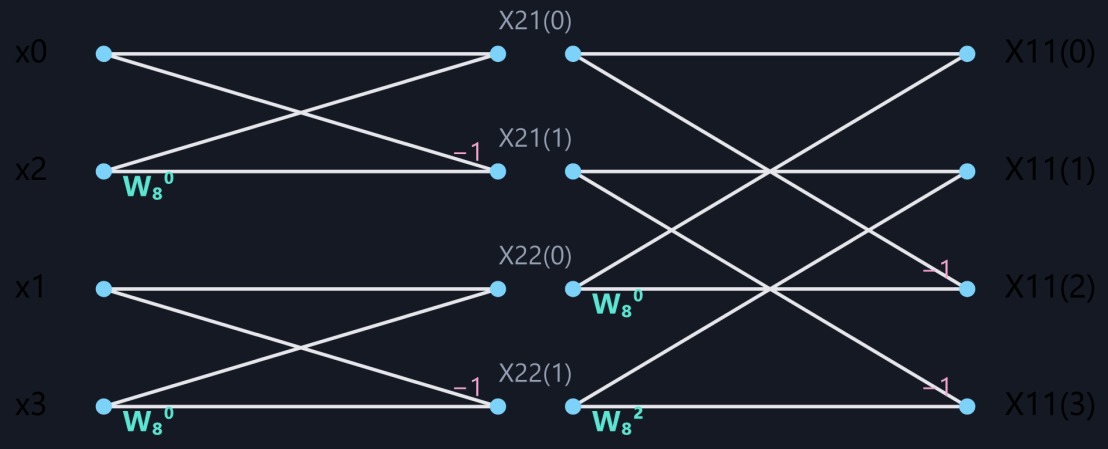
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 0 = 0$$
$$X22(1) = x1 - x3 = 0 - 0 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 0 = 2$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(0) = 0 - j(0) = 0$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 0 = 2$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(0) = 0 + j(0) = 0$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 0$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[0 + j0] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[0 - j0] = \mathbf{0}$$

$\therefore$ x(n) = **{1, 0, 1, 0}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2 + 0]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j2\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(1)]$$
$$= \tfrac{1}{4}\ [2 + 2]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j3\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 0}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

|X(k)| = {2, 0, 2, 0}

$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 0^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + 0^2 + 2^2 + 0^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	0	2	0
|X(k)|	2	0	2	0
Φ(k)	0°	0°	0°	0°

Spectra of {1, 0, 1, 0}



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(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

النتيجة `X(k) = {2, 0, 2, 0} · IDFT → {1, 0, 1, 0} · Parseval 2 = 2 · saving 12 mult., 4 add.`

Key words (iii) Parseval's theorem

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`5-Exams/Final/ · 5-Exams/Midterm/`

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×2 **P11**

x(n) = {1, 1, 0, 1}

For the sequence x(n) = {1, 1, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 1, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 1 + 0 + 1 \\ &= \mathbf{3} \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + e^{-j\pi/2} + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(\pi/2) - j\,\sin(\pi/2)] + [\cos(3\pi/2) - j\,\sin(3\pi/2)] \\ &= 1 + (-j) + j \\ &= \mathbf{1} \end{aligned}$$

Thus X(1) = 1, of magnitude 1 and phase angle Φ(1) = 0°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + e^{-j\pi} + 0 + e^{-j3\pi} \\ &= 1 + [\cos(\pi) - j\,\sin(\pi)] + [\cos(3\pi) - j\,\sin(3\pi)] \\ &= 1 + (-1) + (-1) \\ &= \mathbf{-1} \end{aligned}$$

Thus X(2) = −1, of magnitude 1 and phase angle Φ(2) = 180°.

With k = 3:

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$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + e^{-j3\pi/2} + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\,\sin(3\pi/2)] + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + j + (-j)$$
$$= \mathbf{1}$$

Thus X(3) = 1, of magnitude 1 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 1, 0, 1} has the DFT given by **X(k) = {3, 1, −1, 1}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 1, x2 = 0, x3 = 1, and

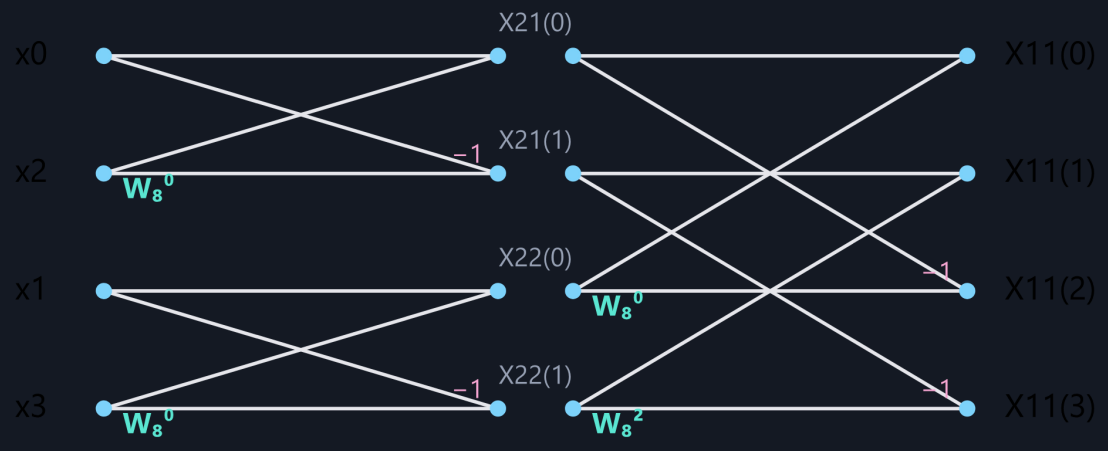
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 1 + 1 = 2$$
$$X22(1) = x1 - x3 = 1 - 1 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 2 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(0) = 1 - j(0) = \mathbf{1}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 2 = \mathbf{-1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(0) = 1 + j(0) = \mathbf{1}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 4$$
$$x22(0) = X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[4 + j0] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[4 - j0] = \mathbf{1}$$

∴ x(n) = **{1, 1, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi/2} + (-1)\ e^{j\pi} + 1\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(j) + (-1)(-1) + 1(-j)]$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi} + (-1)\ e^{j2\pi} + 1\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + 1(-1) + (-1)(1) + 1(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j3\pi/2} + (-1)\ e^{j3\pi} + 1\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(-j) + (-1)(-1) + 1(j)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

$\therefore$ x(n) = **{1, 1, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)|$ = {3, 1, 1, 1}

$$\text{L.H.S.} = 1^2 + 1^2 + 0^2 + 1^2 = \mathbf{3}$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	1	−1	1
$\|X(k)\|$	3	1	1	1
Φ(k)	0°	0°	180°	0°

Spectra of {1, 1, 0, 1}



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(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

النتيجة `X(k) = {3, 1, −1, 1} · IDFT → {1, 1, 0, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.`

Key words (iii) Parseval's theorem

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×3 P12

x(n) = {1, 0, 0, 1}

For the sequence x(n) = {1, 0, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \quad (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 0 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(3\pi/2) - j\, \sin(3\pi/2)] \\ &= 1 + j \\ &= \mathbf{1 + j} \end{aligned}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi} \\ &= 1 + [\cos(3\pi) - j\, \sin(3\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

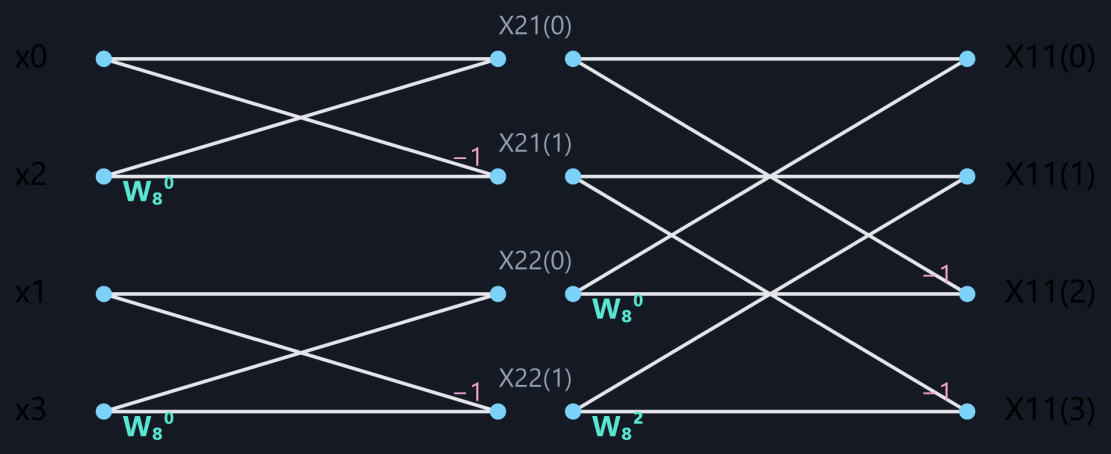
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi/2} + 0 + (1 - j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi} + 0 + (1 - j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)|$ = {2, √2, 0, √2}

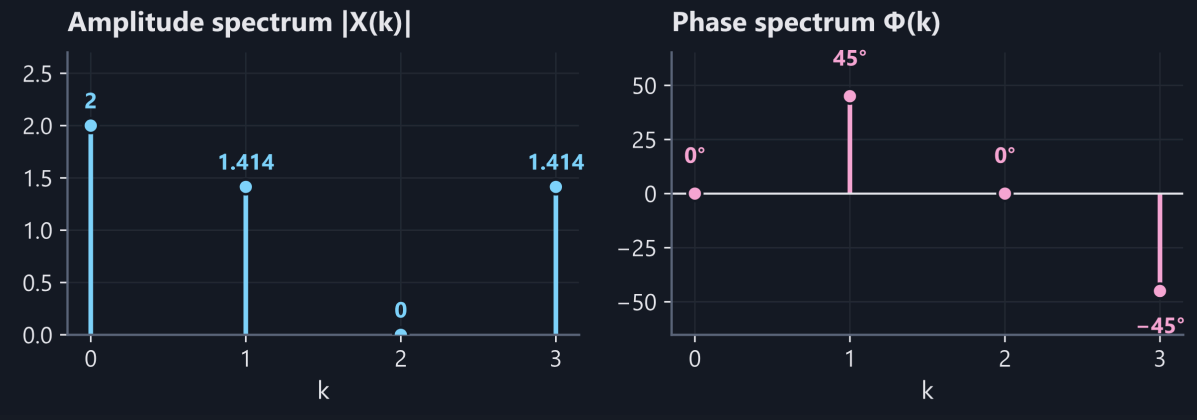
$$\text{L.H.S.} = 1^2 + 0^2 + 0^2 + 1^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
$\|X(k)\|$	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

Spectra of {1, 0, 0, 1}



مسائل الامتحانات القديمة

(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

النتيجة `X(k) = {2, 1 + j, 0, 1 − j} · IDFT → {1, 0, 0, 1} · Parseval 2 = 2 · saving 12 mult., 4 add.`

Key words (iii) Parseval's theorem

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المحاضرة **LEC 5** · Sec. 2.3 · p. 27
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`5-Exams/Final/ · 5-Exams/Midterm/`

★ Cheat sheet · مسائل الامتحانات القديمة

النتايج النهائية

- X(k) = {3, j, 1, −j} · IDFT → {1, 0, 1, 1} · Parseval 3 = 3 · saving 12 mult., 4 add. — **P9**
- X(k) = {2, 0, 2, 0} · IDFT → {1, 0, 1, 0} · Parseval 2 = 2 · saving 12 mult., 4 add. — **P10**
- X(k) = {3, 1, −1, 1} · IDFT → {1, 1, 0, 1} · Parseval 3 = 3 · saving 12 mult., 4 add. — **P11**
- X(k) = {2, 1 + j, 0, 1 − j} · IDFT → {1, 0, 0, 1} · Parseval 2 = 2 · saving 12 mult., 4 add. — **P12**

SECTION 03

Sheet 2

الجزء النظري

الأسئلة

p. 35 Prove that the DFT, X(k), of a sequence x(n) is periodic. 2.1 ☐

2.1

Prove that the DFT, X(k), of a sequence x(n) is periodic.

An important property of the DFT may be deduced if the kth component of the DFT, X(k), is compared with the $(k + N)^{th}$ component, X(k + N). Thus:

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N}$$

$$X(k + N) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi n} = X(k)$$

since n is integral, so $e^{-j2\pi n} = 1$.

The fact that **X(k + N) = X(k)** shows that the DFT is periodic with period N. This is the cyclical property of the DFT: the values of the DFT components are repetitive. If k = 0, then X(0) = X(N); e.g. for {1, 0, 0, 1}, X(0) = 2 and therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



e^{−j2πn} = 1, so X(k + N) = X(k): the DFT is periodic with period N. **الفكرة**

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Sheet 2

Sheet 2 · Q 2.1 الشيت

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Cheat sheet · Sheet 2 ★

افتكر

e^{−j2πn} = 1, so X(k + N) = X(k): the DFT is periodic with period N. 2.1

SECTION 04

Sheet 2

المسائل

الأسئلة

p. 38	DFT of {0, 1, 1, 0} and check by the IDFT	2.2 ☐
p. 41	Verify Parseval's theorem for {0, 1, 1, 0}	2.3 ☐
p. 42	FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT	2.4 ☐
p. 45	FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT	2.5 ☐
p. 49	x(n) = cos(nπ/2): 8 samples, FFT and computational saving	2.6 ☐

2.2

DFT of {0, 1, 1, 0} and check by the IDFT

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

DFT (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1}\, x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1}\, x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \quad (2.4)$$

x(0) = 0, x(T) = 1, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 0 + 1 + 1 + 0 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4} \\ &= 0 + e^{-j\pi/2} + e^{-j\pi} + 0 \\ &= [\cos(\pi/2) - j\,\sin(\pi/2)] + [\cos(\pi) - j\,\sin(\pi)] \\ &= (-j) + (-1) \\ &= \mathbf{-1 - j} \end{aligned}$$

Thus X(1) = −1 − j, of magnitude √2 = 1.414 and phase angle Φ(1) = −135°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4} \\ &= 0 + e^{-j\pi} + e^{-j2\pi} + 0 \\ &= [\cos(\pi) - j\,\sin(\pi)] + [\cos(2\pi) - j\,\sin(2\pi)] \\ &= (-1) + 1 \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 0 + e^{-j3\pi/2} + e^{-j3\pi} + 0$$
$$= [\cos(3\pi/2) - j\,\sin(3\pi/2)] + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= j + (-1)$$
$$= \mathbf{-1 + j}$$

Thus X(3) = −1 + j, of magnitude √2 = 1.414 and phase angle Φ(3) = 135°.

So, the time series x(n) = {0, 1, 1, 0} has the DFT given by **X(k) = {2, −1 − j, 0, −1 + j}**.

IDFT check (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\,[2 + (-1 - j) + 0 + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)\, e^{j\pi/2} + 0 + (-1 + j)\, e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)(j) + (-1 + j)(-j)]$$
$$= \tfrac{1}{4}\,[2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)\, e^{j\pi} + 0 + (-1 + j)\, e^{j3\pi}]$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)(-1) + (-1 + j)(-1)]$$
$$= \tfrac{1}{4}\,[2 + (1 + j) + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)\, e^{j3\pi/2} + 0 + (-1 + j)\, e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\,[2 + (-1 - j)(-j) + (-1 + j)(j)]$$

$$= \tfrac{1}{4}\,[2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{0, 1, 1, 0}**, the original sequence.

النتيجة $X(k) = \{2, -1 - j, 0, -1 + j\} \cdot \text{IDFT} \rightarrow x(n) = \{0, 1, 1, 0\}$

Key words DFT (as Example 2.1)

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2.3

Verify Parseval's theorem for {0, 1, 1, 0}

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0, 1, 1, 0}.

The normalized energy in the signal is given by either of the expressions:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$

$$\text{L.H.S.} = 0^2 + 1^2 + 1^2 + 0^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

النتيجة $\Sigma x^2 = 2 = (1/4)(4 + 2 + 0 + 2) \rightarrow$ Parseval's theorem verified

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2.4

FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT

2.4 Calculate the DFT of the data sequence {0, 1, 1, 0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

FFT (as Example 2.3):

This is a four-point DFT with x0 = 0, x1 = 1, x2 = 1, x3 = 0, and

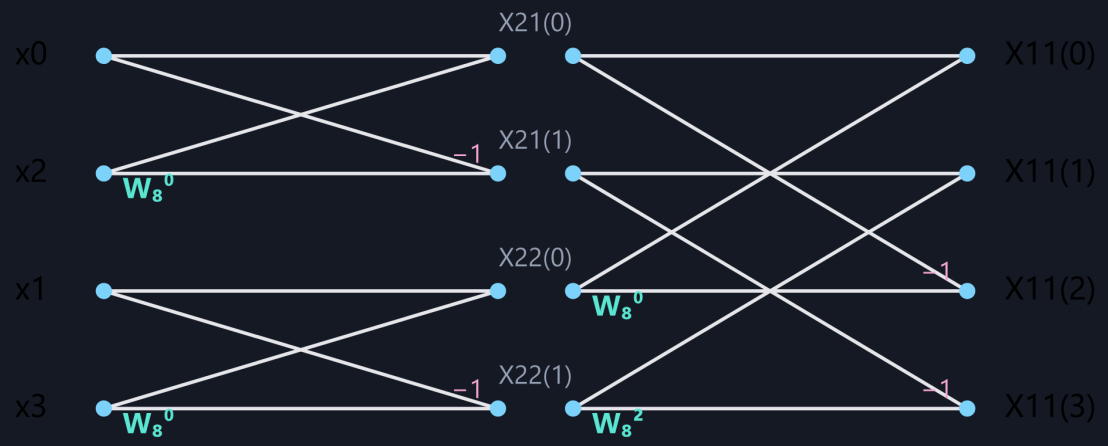
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 0 + 1 = 1$$
$$X21(1) = x0 - x2 = 0 - 1 = -1$$
$$X22(0) = x1 + x3 = 1 + 0 = 1$$
$$X22(1) = x1 - x3 = 1 - 0 = 1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = -1 + e^{-j\pi/2}(1) = -1 - j(1) = \mathbf{-1 - j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = -1 - e^{-j\pi/2}(1) = -1 + j(1) = \mathbf{-1 + j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values X(k) = {2, −1 − j, 0, −1 + j} are the same as those obtained in Prob. (2.2), but were more readily obtained using the FFT algorithm.

Comparison (Sec. 2.4, 2.5.2), N = 4:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = -2 \qquad x22(1) = X(1) - X(3) = -j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\,x22(1)] = \tfrac{1}{4}[2 + j(-j2)] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\,x22(1)] = \tfrac{1}{4}[2 - j(-j2)] = \mathbf{0}$$

∴ x(n) = **{0, 1, 1, 0}**, the original data sequence.

النتيجة FFT gives the same X(k) = {2, −1 − j, 0, −1 + j} · IFFT → x(n) = {0, 1, 1, 0}

Key words FFT (as Example 2.3) IFFT (Sec. 2.6)

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الشيت **Sheet 2** · Q 2.4

Sheet 2

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2.5

FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT

2.5 Calculate the FFT of the data sequence {0, 0, 1, 1, 1, 1, 0, 0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

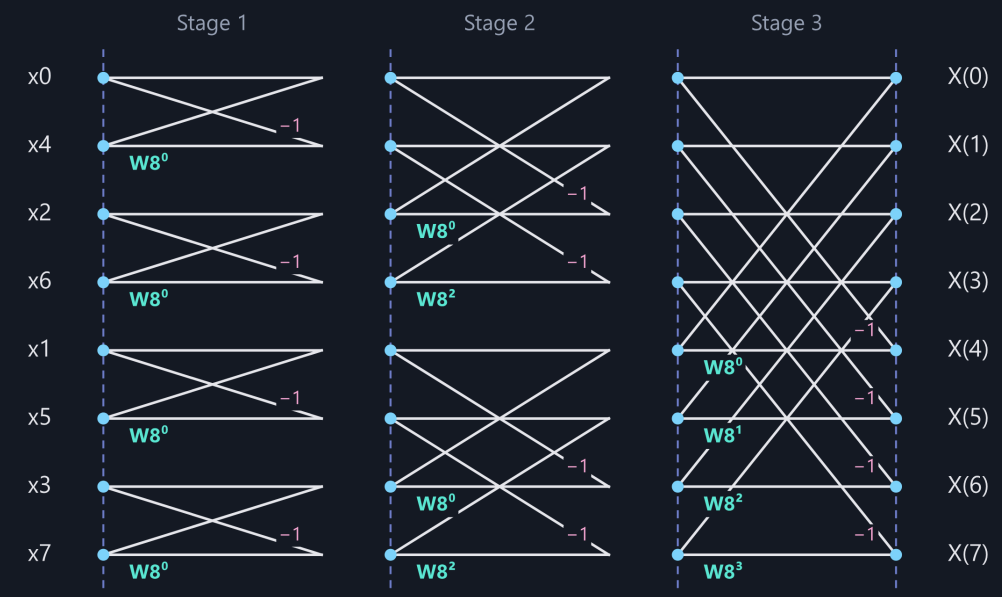
Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 1$ $X21(1) = x0 - x4 = -1$	$X11(0) = X21(0) + W_8^0 X22(0) = 2$ $X11(1) = X21(1) + W_8^2 X22(1) = -1 - j$
$X22(0) = x2 + x6 = 1$ $X22(1) = x2 - x6 = 1$	$X11(2) = X21(0) - W_8^0 X22(0) = 0$ $X11(3) = X21(1) - W_8^2 X22(1) = -1 + j$
$X23(0) = x1 + x5 = 1$ $X23(1) = x1 - x5 = -1$	$X12(0) = X23(0) + W_8^0 X24(0) = 2$ $X12(1) = X23(1) + W_8^2 X24(1) = -1 - j$
$X24(0) = x3 + x7 = 1$ $X24(1) = x3 - x7 = 1$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = -1 + j$

Stage 3: $X1(k) = X11(k) + W_8^k\, X12(k), \quad X1(k+4) = X11(k) - W_8^k\, X12(k), \quad k = 0, 1, 2, 3$

$$X(0) = 4 \qquad X(4) = 0$$
$$X(1) = -2.414 - j \qquad X(5) = 0.414 - j$$
$$X(2) = 0 \qquad X(6) = 0$$
$$X(3) = 0.414 + j \qquad X(7) = -2.414 + j$$

$\therefore X(k) =$ **{4, −2.414 − j, 0, 0.414 + j, 0, 0.414 − j, 0, −2.414 + j}**

8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)

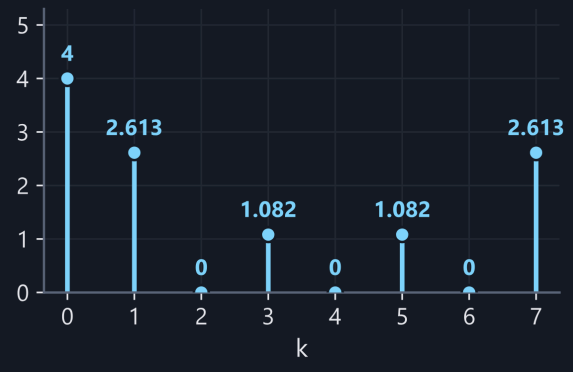


Amplitude and phase spectra ($|X(k)|$ and $\Phi(k)$ versus k; symmetrical about N/2 = 4):

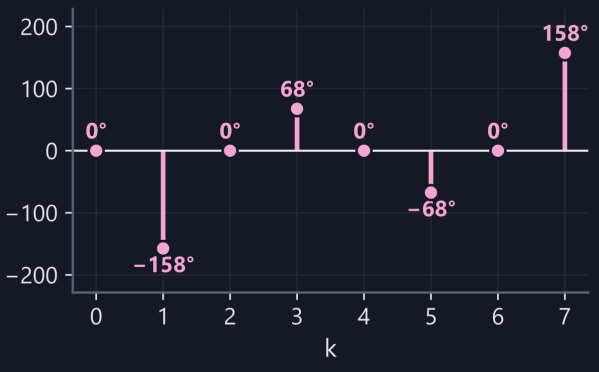
k	0	1	2	3	4	5	6	7
X(k)	4	−2.414 − j	0	0.414 + j	0	0.414 − j	0	−2.414 + j
$\|X(k)\|$	4	2.613	0	1.082	0	1.082	0	2.613
$\Phi(k)$	0°	−157.5°	0°	67.5°	0°	−67.5°	0°	157.5°

Spectra of {0, 0, 1, 1, 1, 1, 0, 0}

Amplitude spectrum |X(k)|



Phase spectrum Φ(k)



IFFT check (same flowgraph, W_8^{-k}, divided by N = 8):

Re-ordered input (Table 2.1): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7). $W_8^{0} = 1$, $W_8^{-1} = (1 + j)/\sqrt{2}$, $W_8^{-2} = +j$, $W_8^{-3} = (-1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = X(0) + X(4) = 4$ $X21(1) = X(0) - X(4) = 4$	$X11(0) = X21(0) + W_8^{0}X22(0) = 4$ $X11(1) = X21(1) + W_8^{-2}X22(1) = 4$
$X22(0) = X(2) + X(6) = 0$ $X22(1) = X(2) - X(6) = 0$	$X11(2) = X21(0) - W_8^{0}X22(0) = 4$ $X11(3) = X21(1) - W_8^{-2}X22(1) = 4$
$X23(0) = X(1) + X(5) = -2 - j2$ $X23(1) = X(1) - X(5) = -2.828$	$X12(0) = X23(0) + W_8^{0}X24(0) = -4$ $X12(1) = X23(1) + W_8^{-2}X24(1) = -2.828 + j2.828$
$X24(0) = X(3) + X(7) = -2 + j2$ $X24(1) = X(3) - X(7) = 2.828$	$X12(2) = X23(0) - W_8^{0}X24(0) = -j4$ $X12(3) = X23(1) - W_8^{-2}X24(1) = -2.828 - j2.828$

Stage 3: $X1(k) = X11(k) + W_8^{-k}\, X12(k), \quad X1(k+4) = X11(k) - W_8^{-k}\, X12(k), \quad k = 0, 1, 2, 3$

$x(0) = (1/8)(0) = \mathbf{0}$ $\quad$ $x(4) = (1/8)(8) = \mathbf{1}$

$x(1) = (1/8)(0) = \mathbf{0}$ $\quad$ $x(5) = (1/8)(8) = \mathbf{1}$

$x(2) = (1/8)(8) = \mathbf{1}$ $\quad$ $x(6) = (1/8)(0) = \mathbf{0}$

$x(3) = (1/8)(8) = \mathbf{1}$ $\quad$ $x(7) = (1/8)(0) = \mathbf{0}$

$\therefore$ x(n) = **{0, 0, 1, 1, 1, 1, 0, 0}**, the original data sequence.

النتيجة $X(k) = \{4, -2.414 - j, 0, 0.414 + j, 0, 0.414 - j, 0, -2.414 + j\}$ · $|X(k)| = \{4, 2.613, 0, 1.082, 0, 1.082, 0, 2.613\}$ · IFFT → {0, 0, 1, 1, 1, 1, 0, 0}

Key words FFT (8-point, Fig. 2.3)

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 6** · Table 2.1 · p. 10
1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-6-Ch-2.pdf

المحاضرة **LEC 6** · Sec. 2.6 · p. 20
1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-6-Ch-2.pdf

Sheet 2

Sheet 2 · Q 2.5 الشيت

2-Section/Sheets/Sheet 2.pdf · solved: 2-Section/Sheets/Solved Sheet/Sheet 2 Solved.pdf

2.6

x(n) = cos(nπ/2): 8 samples, FFT and computational saving

2.6 For the signal x(n) = cos(nπ/2), find the first 8 samples and then, using the FFT algorithm, calculate the frequency components X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving?

$$x(n) = \cos(n\pi/2): \quad x(0) = \cos 0 = 1,\ x(1) = \cos(\pi/2) = 0,\ x(2) = \cos \pi = -1,\ x(3) = \cos(3\pi/2) = 0,\ \ldots$$
$$x(n) = \{1, 0, -1, 0, 1, 0, -1, 0\}$$

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
X21(0) = x0 + x4 = 2 X21(1) = x0 − x4 = 0	X11(0) = X21(0) + W_8^0X22(0) = 0 X11(1) = X21(1) + W_8^2X22(1) = 0
X22(0) = x2 + x6 = −2 X22(1) = x2 − x6 = 0	X11(2) = X21(0) − W_8^0X22(0) = 4 X11(3) = X21(1) − W_8^2X22(1) = 0
X23(0) = x1 + x5 = 0 X23(1) = x1 − x5 = 0	X12(0) = X23(0) + W_8^0X24(0) = 0 X12(1) = X23(1) + W_8^2X24(1) = 0
X24(0) = x3 + x7 = 0 X24(1) = x3 − x7 = 0	X12(2) = X23(0) − W_8^0X24(0) = 0 X12(3) = X23(1) − W_8^2X24(1) = 0

Stage 3: $X1(k) = X11(k) + W_8^k\, X12(k), \quad X1(k + 4) = X11(k) - W_8^k\, X12(k), \quad k = 0, 1, 2, 3$

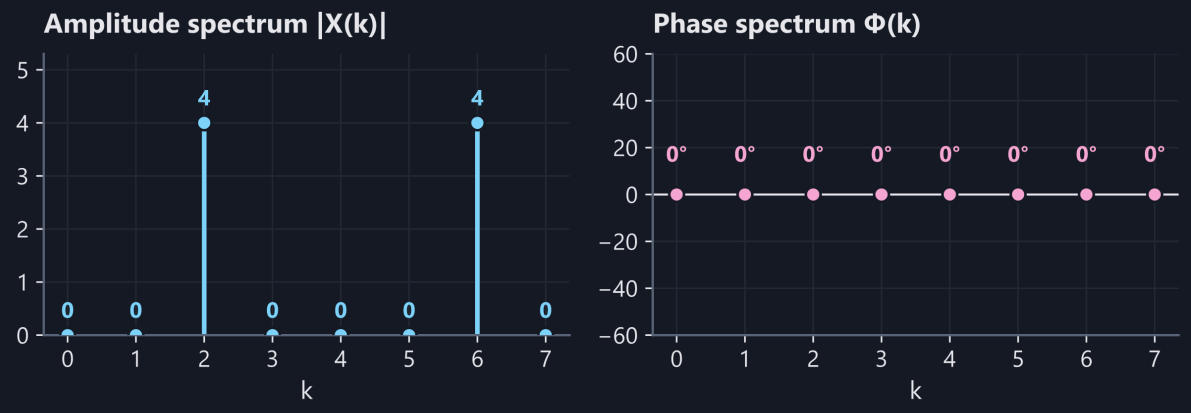
X(0) = 0	X(4) = 0
X(1) = 0	X(5) = 0
X(2) = 4	X(6) = 4
X(3) = 0	X(7) = 0

∴ X(k) = **{0, 0, 4, 0, 0, 0, 4, 0}**

k	0	1	2	3	4	5	6	7

X(k)	0	0	4	0	0	0	4	0
|X(k)|	0	0	4	0	0	0	4	0
Φ(k)	0°	0°	0°	0°	0°	0°	0°	0°

Spectra of x(n) = cos(nπ/2)



Computational saving (Sec. 2.5.2), N = 8:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 64$	$(N/2)\log_2 N = 12$	**52**
No. of complex additions	$N(N-1) = 56$	$N\log_2 N = 24$	**32**

النتيجة `x(n) = {1, 0, −1, 0, 1, 0, −1, 0} · X(k) = {0, 0, 4, 0, 0, 0, 4, 0} · saving 52 mult., 32 add.`

Key words FFT (8-point, Fig. 2.3)

المصدر — من الـ PDFs بتاعتك

LEC 6 · Sec. 2.5 · p. 6 — المحاضرة
`1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-6-Ch-2.pdf`

LEC 6 · Sec. 2.5.2 · p. 18 — المحاضرة
`1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-6-Ch-2.pdf`

Sheet 2

Sheet 2 · Q 2.6 — الشيت

2-Section/Sheets/Sheet 2.pdf · solved: 2-Section/Sheets/Solved Sheet/Sheet 2 Solved.pdf

Cheat sheet · Sheet 2 ★

النتايج النهائية

$X(k) = \{2, -1 - j, 0, -1 + j\}$ · IDFT → $x(n) = \{0, 1, 1, 0\}$	2.2
$\Sigma x^2 = 2 = (1/4)(4 + 2 + 0 + 2)$ → Parseval's theorem verified	2.3
FFT gives the same $X(k) = \{2, -1 - j, 0, -1 + j\}$ · IFFT → $x(n) = \{0, 1, 1, 0\}$	2.4
$X(k) = \{4, -2.414 - j, 0, 0.414 + j, 0, 0.414 - j, 0, -2.414 + j\}$ · $\lvert X(k)\rvert = \{4, 2.613, 0, 1.082, 0, 1.082, 0, 2.613\}$ · IFFT → $\{0, 0, 1, 1, 1, 1, 0, 0\}$	2.5
$x(n) = \{1, 0, -1, 0, 1, 0, -1, 0\}$ · $X(k) = \{0, 0, 4, 0, 0, 0, 4, 0\}$ · saving 52 mult., 32 add.	2.6

SECTION 05

حل الـ Assignment

DFT / IDFT / FFT / IFFT

الأسئلة

p. 53 DSP Lecture Assignment — Fall 27 Q ☐

Q

DSP Lecture Assignment — Fall 27

*For the sequence **x(n) = {0, 2, 1, 1, 3, 1, 2, 0}**, calculate the frequency components X(k) by the DFT method and check the results by calculating x(n) by the IDFT method. Repeat the calculation of X(k) by the FFT algorithm and check by calculating x(n) by the IFFT algorithm.*

(1) Frequency components X(k) by the DFT method

The DFT of an N-point sequence is given by Eq. (2.4):

$$X(k) = F_D[x(nT)] = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \qquad (2.4)$$
$$\text{with } \Omega = 2\pi/NT, \text{ so that } k\Omega nT = 2\pi kn/N$$

Here N = 8 and x(0) = 0, x(T) = 2, x(2T) = 1, x(3T) = 1, x(4T) = 3, x(5T) = 1, x(6T) = 2, x(7T) = 0. It is required to find X(k) for k = 0, 1, ..., 7. The exponent is $-j2\pi kn/8 = -j\pi kn/4$; the term n = 0 vanishes because x(0) = 0. Values used: $e^{\mp j\pi/4} = 0.707 \mp j0.707$, $e^{\mp j\pi/2} = \mp j$, $e^{\mp j3\pi/4} = -0.707 \mp j0.707$, $e^{\mp j\pi} = -1$.

With k = 0:

$$X(0) = \sum_{n=0}^{7} x(nT)\, e^{-j2\pi(0)n/8} = \sum_{n=0}^{7} x(nT)\, e^{-j0\cdot n}$$
$$= 0 + 2 + 1 + 1 + 3 + 1 + 2$$
$$= 2 + 1 + 1 + 3 + 1 + 2$$
$$= 2 + 1 + 1 + 3 + 1 + 2$$
$$= \mathbf{10}$$

Thus X(0) = 10, of magnitude 10 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \sum_{n=0}^{7} x(nT)\, e^{-j2\pi(1)n/8} = \sum_{n=0}^{7} x(nT)\, e^{-j\pi/4\cdot n}$$
$$= 0 + 2e^{-j\pi/4} + e^{-j\pi/2} + e^{-j3\pi/4} + 3e^{-j\pi} + e^{j3\pi/4} + 2e^{j\pi/2}$$
$$= 2(0.707 - j0.707) + (-j) + (-0.707 - j0.707) + 3(-1) + (-0.707 + j0.707) + 2(j)$$
$$= (1.414 - j1.414) + (-j) + (-0.707 - j0.707) + (-3) + (-0.707 + j0.707) + j2$$
$$= \mathbf{-3 - j0.414}$$

Thus X(1) = −3 − j0.414, of magnitude 3.028 and phase angle Φ(1) = −172.1°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(2)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j\pi/2\cdot n}$$
$$= 0 + 2e^{-j\pi/2} + e^{-j\pi} + e^{j\pi/2} + 3 + e^{-j\pi/2} + 2e^{-j\pi}$$
$$= 2(-j) + (-1) + (j) + 3 + (-j) + 2(-1)$$
$$= (-j2) + (-1) + j + 3 + (-j) + (-2)$$
$$= \mathbf{-j2}$$

Thus X(2) = –j2, of magnitude 2 and phase angle Φ(2) = −90°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(3)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j3\pi/4\cdot n}$$
$$= 0 + 2e^{-j3\pi/4} + e^{j\pi/2} + e^{-j\pi/4} + 3e^{-j\pi} + e^{j\pi/4} + 2e^{-j\pi/2}$$
$$= 2(-0.707 - j0.707) + (j) + (0.707 - j0.707) + 3(-1) + (0.707 + j0.707) + 2(-j)$$
$$= (-1.414 - j1.414) + j + (0.707 - j0.707) + (-3) + (0.707 + j0.707) + (-j2)$$
$$= \mathbf{-3 - j2.414}$$

Thus X(3) = −3 − j2.414, of magnitude 3.851 and phase angle Φ(3) = −141.2°.

With k = 4:

$$X(4) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(4)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j\pi\cdot n}$$
$$= 0 + 2e^{-j\pi} + 1 + e^{-j\pi} + 3 + e^{-j\pi} + 2$$
$$= 2(-1) + 1 + (-1) + 3 + (-1) + 2$$
$$= (-2) + 1 + (-1) + 3 + (-1) + 2$$
$$= \mathbf{2}$$

Thus X(4) = 2, of magnitude 2 and phase angle Φ(4) = 0°.

With k = 5:

$$X(5) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(5)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j-3\pi/4\cdot n}$$
$$= 0 + 2e^{j3\pi/4} + e^{-j\pi/2} + e^{j\pi/4} + 3e^{-j\pi} + e^{-j\pi/4} + 2e^{j\pi/2}$$
$$= 2(-0.707 + j0.707) + (-j) + (0.707 + j0.707) + 3(-1) + (0.707 - j0.707) + 2(j)$$
$$= (-1.414 + j1.414) + (-j) + (0.707 + j0.707) + (-3) + (0.707 - j0.707) + j2$$
$$= \mathbf{-3 + j2.414}$$

Thus X(5) = −3 + j2.414, of magnitude 3.851 and phase angle Φ(5) = 141.2°.

With k = 6:

$$X(6) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(6)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j-\pi/2\cdot n}$$
$$= 0 + 2e^{j\pi/2} + e^{-j\pi} + e^{-j\pi/2} + 3 + e^{j\pi/2} + 2e^{-j\pi}$$

$$= 2(j) + (-1) + (-j) + 3 + (j) + 2(-1)$$
$$= j2 + (-1) + (-j) + 3 + j + (-2)$$
$$= \mathbf{j2}$$

Thus X(6) = j2, of magnitude 2 and phase angle Φ(6) = 90°.

With k = 7:

$$X(7) = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j2\pi(7)n/8} = \Sigma_{n=0}^{7}\, x(nT)\, e^{-j-\pi/4\cdot n}$$
$$= 0 + 2e^{j\pi/4} + e^{j\pi/2} + e^{j3\pi/4} + 3e^{-j\pi} + e^{-j3\pi/4} + 2e^{-j\pi/2}$$
$$= 2(0.707 + j0.707) + (j) + (-0.707 + j0.707) + 3(-1) + (-0.707 - j0.707) + 2(-j)$$
$$= (1.414 + j1.414) + j + (-0.707 + j0.707) + (-3) + (-0.707 - j0.707) + (-j2)$$
$$= \mathbf{-3 + j0.414}$$

Thus X(7) = −3 + j0.414, of magnitude 3.028 and phase angle Φ(7) = 172.1°.

So, the time series x(n) = {0, 2, 1, 1, 3, 1, 2, 0} has the DFT

$$\mathbf{X(k) = \{10, -3 - j0.414, -j2, -3 - j2.414, 2, -3 + j2.414, j2, -3 + j0.414\}}$$

k	0	1	2	3	4	5	6	7
X(k)	10	−3 − j0.414	−j2	−3 − j2.414	2	−3 + j2.414	j2	−3 + j0.414
|X(k)|	10	3.028	2	3.851	2	3.851	2	3.028
Φ(k)	0°	−172.1°	−90°	−141.2°	0°	141.2°	90°	172.1°

Note: X(8 − k) = X*(k) (e.g. X(7) = X*(1)), as expected for a real sequence.

(2) Check: original sequence x(n) by the IDFT method

The IDFT is given by Eq. (2.7):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\, \Sigma_{k=0}^{N-1}\, X(k)\, e^{jk\Omega nT} = (1/8)\, \Sigma_{k=0}^{7}\, X(k)\, e^{j\pi kn/4} \qquad (2.7)$$

With n = 0:

$$x(0T) = (1/8)\, \Sigma_{k=0}^{7}\, X(k)\, e^{j2\pi k(0)/8}$$
$$= (1/8)[10 + (-3 - j0.414) + (-j2) + (-3 - j2.414) + 2 + (-3 + j2.414) + j2 + (-3 + j0.414)]$$
$$= (1/8)(0) = \mathbf{0}$$

With n = 1:

$$x(1T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(1)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi/4} + (-j2)e^{j\pi/2} + (-3 - j2.414)e^{j3\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{-j3\pi/4} + j2e^{-j\pi/2} + (-3 + j0.414)e^{-j\pi/4}]$$
$$= (1/8)[10 + (-1.828 - j2.414) + 2 + (3.828 - j0.414) + (-2) + (3.828 + j0.414) + 2 + (-1.828 + j2.414)]$$
$$= (1/8)(16) = \mathbf{2}$$

With n = 2:

$$x(2T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(2)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi/2} + (-j2)e^{j\pi} + (-3 - j2.414)e^{-j\pi/2} + 2 + (-3 + j2.414)e^{j\pi/2} + j2e^{j\pi} + (-3 + j0.414)e^{-j\pi/2}]$$
$$= (1/8)[10 + (0.414 - j3) + j2 + (-2.414 + j3) + 2 + (-2.414 - j3) + (-j2) + (0.414 + j3)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 3:

$$x(3T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(3)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j3\pi/4} + (-j2)e^{-j\pi/2} + (-3 - j2.414)e^{j\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{-j\pi/4} + j2e^{j\pi/2} + (-3 + j0.414)e^{-j3\pi/4}]$$
$$= (1/8)[10 + (2.414 - j1.828) + (-2) + (-0.414 - j3.828) + (-2) + (-0.414 + j3.828) + (-2) + (2.414 + j1.828)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 4:

$$x(4T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(4)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi} + (-j2) + (-3 - j2.414)e^{j\pi} + 2 + (-3 + j2.414)e^{j\pi} + j2 + (-3 + j0.414)e^{j\pi}]$$
$$= (1/8)[10 + (3 + j0.414) + (-j2) + (3 + j2.414) + 2 + (3 - j2.414) + j2 + (3 - j0.414)]$$
$$= (1/8)(24) = \mathbf{3}$$

With n = 5:

$$x(5T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(5)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j3\pi/4} + (-j2)e^{j\pi/2} + (-3 - j2.414)e^{-j\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{j\pi/4} + j2e^{-j\pi/2} + (-3 + j0.414)e^{j3\pi/4}]$$
$$= (1/8)[10 + (1.828 + j2.414) + 2 + (-3.828 + j0.414) + (-2) + (-3.828 -$$

$$j0.414) + 2 + (1.828 - j2.414)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 6:

$$x(6T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(6)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j\pi/2} + (-j2)e^{j\pi} + (-3 - j2.414)e^{j\pi/2} + 2 + (-3 + j2.414)e^{-j\pi/2} + j2e^{j\pi} + (-3 + j0.414)e^{j\pi/2}]$$
$$= (1/8)[10 + (-0.414 + j3) + j2 + (2.414 - j3) + 2 + (2.414 + j3) + (-j2) + (-0.414 - j3)]$$
$$= (1/8)(16) = \mathbf{2}$$

With n = 7:

$$x(7T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(7)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j\pi/4} + (-j2)e^{-j\pi/2} + (-3 - j2.414)e^{-j3\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{j3\pi/4} + j2e^{j\pi/2} + (-3 + j0.414)e^{j\pi/4}]$$
$$= (1/8)[10 + (-2.414 + j1.828) + (-2) + (0.414 + j3.828) + (-2) + (0.414 - j3.828) + (-2) + (-2.414 - j1.828)]$$
$$= (1/8)(0) = \mathbf{0}$$

∴ x(n) = **{0, 2, 1, 1, 3, 1, 2, 0}**, which is the original data sequence, so the DFT results are correct.

(3) Repeat: X(k) by the FFT algorithm (decimation in time)

For N = 8 = 2^3 the DFT is decomposed into $\log_2 8 = 3$ stages: the 8-point DFT X1(k) is expressed by Eq. (2.24) in terms of the 4-point DFTs X11(k) (even samples) and X12(k) (odd samples), and each of these in terms of 2-point DFTs X21(k) ... X24(k), as in Table 2.1.

$$X1(k) = X11(k) + W_N^{\,k}\ X12(k), \quad W_N = e^{-j2\pi/N} \qquad (2.24)$$

Re-ordering (Table 2.1, line 6): x0, x4, x2, x6, x1, x5, x3, x7, where x0 = 0, x1 = 2, x2 = 1, x3 = 1, x4 = 3, x5 = 1, x6 = 2, x7 = 0.

Weights: $W_8^{\,0} = 1$, $W_8^{\,1} = e^{-j\pi/4} = 0.707 - j0.707$, $W_8^{\,2} = -j$, $W_8^{\,3} = e^{-j3\pi/4} = -0.707 - j0.707$.

Stage 1 — four 2-point DFTs (the weight is $W_8^{\,0} = 1$):

$$X21(0) = x0 + x4 = 3, \quad X21(1) = x0 - x4 = -3$$
$$X22(0) = x2 + x6 = 3, \quad X22(1) = x2 - x6 = -1$$

$$X23(0) = x1 + x5 = 3, \quad X23(1) = x1 - x5 = 1$$
$$X24(0) = x3 + x7 = 1, \quad X24(1) = x3 - x7 = 1$$

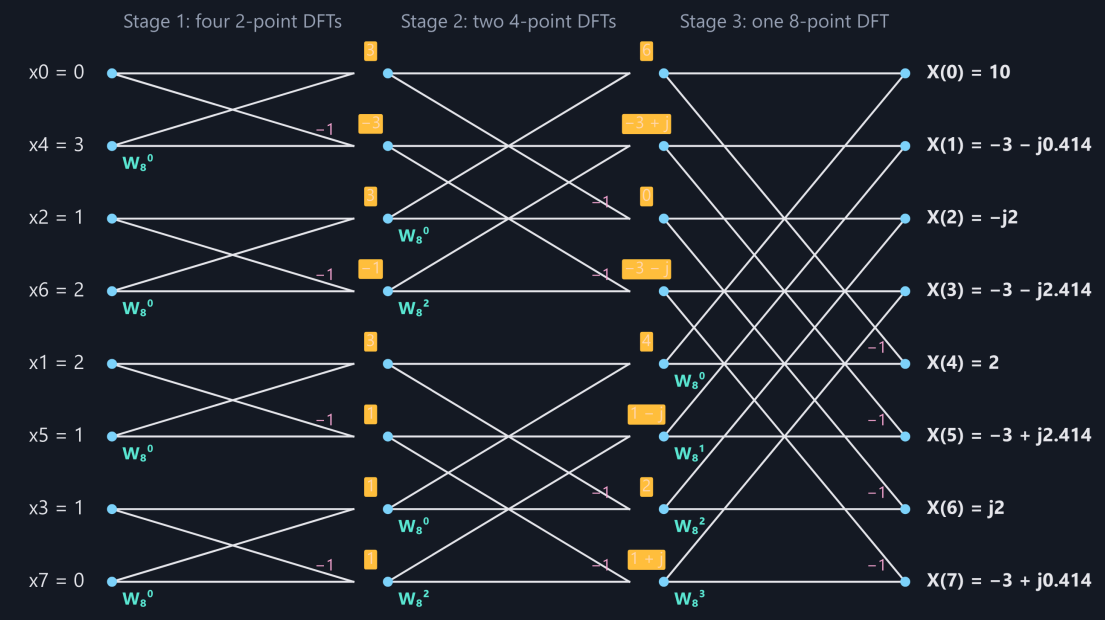
Stage 2 — two 4-point DFTs, Eqs. (2.25a–d), with $W_8^2 = e^{-j\pi/2} = -j$:

$$X11(0) = X21(0) + W_8^0\, X22(0) = 3 + 3 = 6$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = (-3) + (-j)(-1) = -3 + j$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 3 - 3 = 0$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = (-3) - (-j)(-1) = -3 - j$$

$$X12(0) = X23(0) + W_8^0\, X24(0) = 3 + 1 = 4$$
$$X12(1) = X23(1) + W_8^2\, X24(1) = 1 + (-j)1 = 1 - j$$
$$X12(2) = X23(0) - W_8^0\, X24(0) = 3 - 1 = 2$$
$$X12(3) = X23(1) - W_8^2\, X24(1) = 1 - (-j)1 = 1 + j$$

Stage 3 — the 8-point DFT, Eq. (2.24): $X1(k) = X11(k) + W_8^k\, X12(k)$, and $X1(k+4) = X11(k) - W_8^k\, X12(k)$, $k = 0, 1, 2, 3$:

$$X1(0) = X11(0) + W_8^0\, X12(0) = 6 + 14 = 6 + 4 = \mathbf{10}$$
$$X1(1) = X11(1) + W_8^1\, X12(1) = (-3 + j) + (0.707 - j0.707)(1 - j) = (-3 + j) + (-j1.414) = \mathbf{-3 - j0.414}$$
$$X1(2) = X11(2) + W_8^2\, X12(2) = 0 + (-j)2 = 0 + (-j2) = \mathbf{-j2}$$
$$X1(3) = X11(3) + W_8^3\, X12(3) = (-3 - j) + (-0.707 - j0.707)(1 + j) = (-3 - j) + (-j1.414) = \mathbf{-3 - j2.414}$$
$$X1(4) = X11(0) - W_8^0\, X12(0) = 6 - 4 = \mathbf{2}$$
$$X1(5) = X11(1) - W_8^1\, X12(1) = (-3 + j) - (-j1.414) = \mathbf{-3 + j2.414}$$
$$X1(6) = X11(2) - W_8^2\, X12(2) = 0 - (-j2) = \mathbf{j2}$$
$$X1(7) = X11(3) - W_8^3\, X12(3) = (-3 - j) - (-j1.414) = \mathbf{-3 + j0.414}$$



∴ X(k) = **{10, −3 − j0.414, −j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414}** — the same values obtained by the DFT method.

(4) Check: x(n) by the IFFT algorithm

Comparing Eq. (2.27) with Eq. (2.26), the IFFT differs from the FFT only in the factor 1/N and the sign of the exponent. So the same flowgraph is used with input X(k) (re-ordered), weights W_8^{-k} instead of W_8^{k}, and the outputs are divided by N = 8:

$$x(\mu) = (1/N) \sum_{\lambda=0}^{N-1} X(\lambda)\, e^{+j\mu(2\pi/N)\lambda}, \quad \mu = 0, 1, \ldots, N-1 \qquad (2.27)$$

Re-ordering (Table 2.1, line 6): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7), where X(0) = 10, X(1) = −3 − j0.414, X(2) = −j2, X(3) = −3 − j2.414, X(4) = 2, X(5) = −3 + j2.414, X(6) = j2, X(7) = −3 + j0.414.

Weights: $W_8^{0} = 1$, $W_8^{-1} = e^{j\pi/4} = 0.707 + j0.707$, $W_8^{-2} = +j$, $W_8^{-3} = e^{j3\pi/4} = -0.707 + j0.707$.

Stage 1 — four 2-point DFTs (the weight is $W_8^{0} = 1$):

$$X21(0) = X(0) + X(4) = 12, \quad X21(1) = X(0) - X(4) = 8$$
$$X22(0) = X(2) + X(6) = 0, \quad X22(1) = X(2) - X(6) = -j4$$
$$X23(0) = X(1) + X(5) = -6 + j2, \quad X23(1) = X(1) - X(5) = -j2.828$$
$$X24(0) = X(3) + X(7) = -6 - j2, \quad X24(1) = X(3) - X(7) = -j2.828$$

Stage 2 — two 4-point DFTs, Eqs. (2.25a–d), with $W_8^{-2} = e^{+j\pi/2} = +j$:

حل الـ Assignment

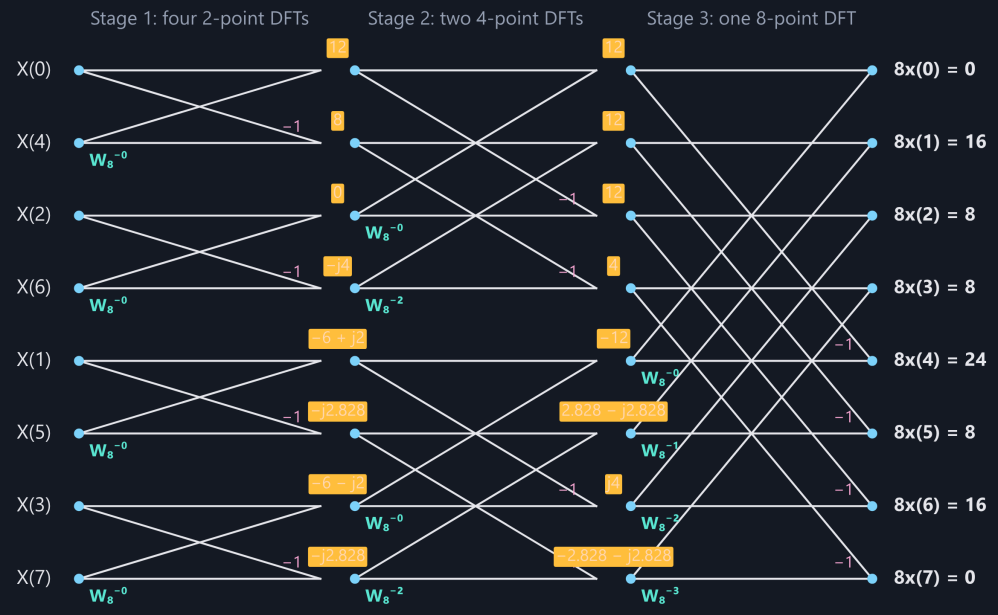
$$X11(0) = X21(0) + W_8^0\, X22(0) = 12 + 0 = 12$$
$$X11(1) = X21(1) + W_8^{-2}\, X22(1) = 8 + (j)(-j4) = 12$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 12 - 0 = 12$$
$$X11(3) = X21(1) - W_8^{-2}\, X22(1) = 8 - (j)(-j4) = 4$$

$$X12(0) = X23(0) + W_8^0\, X24(0) = (-6 + j2) + (-6 - j2) = -12$$
$$X12(1) = X23(1) + W_8^{-2}\, X24(1) = (-j2.828) + (j)(-j2.828) = 2.828 - j2.828$$
$$X12(2) = X23(0) - W_8^0\, X24(0) = (-6 + j2) - (-6 - j2) = j4$$
$$X12(3) = X23(1) - W_8^{-2}\, X24(1) = (-j2.828) - (j)(-j2.828) = -2.828 - j2.828$$

Stage 3 — the 8-point DFT, Eq. (2.24): $X1(k) = X11(k) + W_8^{-k}\, X12(k)$, and $X1(k + 4) = X11(k) - W_8^{-k}\, X12(k)$, $k = 0, 1, 2, 3$:

$$X1(0) = X11(0) + W_8^{-0}\, X12(0) = 12 + 1(-12) = 12 + (-12) = \mathbf{0}$$
$$X1(1) = X11(1) + W_8^{-1}\, X12(1) = 12 + (0.707 + j0.707)(2.828 - j2.828) = 12 + 4 = \mathbf{16}$$
$$X1(2) = X11(2) + W_8^{-2}\, X12(2) = 12 + jj4 = 12 + (-4) = \mathbf{8}$$
$$X1(3) = X11(3) + W_8^{-3}\, X12(3) = 4 + (-0.707 + j0.707)(-2.828 - j2.828) = 4 + 4 = \mathbf{8}$$
$$X1(4) = X11(0) - W_8^{-0}\, X12(0) = 12 - (-12) = \mathbf{24}$$
$$X1(5) = X11(1) - W_8^{-1}\, X12(1) = 12 - 4 = \mathbf{8}$$
$$X1(6) = X11(2) - W_8^{-2}\, X12(2) = 12 - (-4) = \mathbf{16}$$
$$X1(7) = X11(3) - W_8^{-3}\, X12(3) = 4 - 4 = \mathbf{0}$$

IFFT flowgraph: same butterflies, W_8^{-k}, outputs divided by N = 8



حل الـ Assignment

$$x(0) = (1/8)(0) = 0$$
$$x(1) = (1/8)(16) = 2$$
$$x(2) = (1/8)(8) = 1$$
$$x(3) = (1/8)(8) = 1$$
$$x(4) = (1/8)(24) = 3$$
$$x(5) = (1/8)(8) = 1$$
$$x(6) = (1/8)(16) = 2$$
$$x(7) = (1/8)(0) = 0$$

∴ x(n) = **{0, 2, 1, 1, 3, 1, 2, 0}**, the original data sequence, so the FFT results are correct.

Final answer

X(k) = {10, −3 − j0.414, –j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414}

DFT and FFT give the same X(k); IDFT and IFFT both return x(n) = {0, 2, 1, 1, 3, 1, 2, 0}

النتيجة

X(k) = {10, −3 − j0.414, −j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414} · IDFT / IFFT → {0, 2, 1, 1, 3, 1, 2, 0}

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المحاضرة

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